Shivam Govind Rao

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Summary

Full-stack engineer with 2.5 years shipping AI products end to end: React and Next.js frontends, Node.js and Python backends, AWS pipelines and LLM integrations. Owned the backend and both production frontends of Diagna AIś HIPAA-compliant clinical platform (1,000+ documents/day) and took VoiceGenie, a generative-AI voice platform, from 0 to $10K MRR in 11 months. Created PayOps AI, a LangGraph multi-agent app with human approvals.

Experience

Software Development Engineer **Jan 2025 - Apr 2026**
Diagna AI | *diagna.ai* *Gurugram, India*

- Architected and owned the core backend of a HIPAA-compliant clinical document platform serving live paying clinics: **40+ REST endpoints** (TypeScript, Express, JWT + RBAC, Swagger) over a 25-model PostgreSQL/Prisma schema the team built against.
- Delivered both production React 19 frontends: the FAXFlo clinical inbox with a dual-view AI document editor (Zustand + Immer diff-based PATCH editing, PDF.js), and an internal dashboard for merging duplicate patient records with pre-merge validation.
- Engineered the distributed AWS ingestion pipeline (S3, SQS, SNS) with **7 BullMQ/Redis queues**, DB-level idempotency and a dead-letter queue, processing **1,000+ medical documents/day** and self-healing transient failures.
- Integrated Claude, Amazon Nova and Llama via AWS Bedrock to classify documents across 40+ categories with structured JSON extraction, model fallback and retries, cutting manual fax triage from **about 3 hours to 10 minutes a day** and clearing a recurring 2-day backlog.
- Built EHR write-back for a no-API legacy EHR (MDLand) via an RPA service (Python, FastAPI, Robocorp, Playwright, Redis/RQ) plus Medplum FHIR sync, cutting referral turnaround from **4.2 days to same day**.
- Built an AI SMS orchestrator over Twilio, Telnyx and Vonage with queue-backed webhooks and TCPA-style consent; covered critical paths with Jest and Supertest.

Software Engineer **Dec 2023 - Jan 2025**
Oriserve | *voicegenie.ai* *Noida, India*

- Shipped the entire frontend of VoiceGenie (Next.js, React, TypeScript): customer dashboard, AI campaign builder, onboarding and post-call analytics; the product grew from 0 to **$10K MRR in 11 months** and closed its first enterprise customers.
- Integrated Stripe payments end to end (checkout UI, backend APIs, transaction-state handling) and synced HubSpot, GoHighLevel and Cal.com so AI call outcomes updated CRM records and booked meetings automatically.
- Cut page load time by **30%** (LCP, FCP, TBT, CLS) through Lighthouse profiling, code splitting, lazy loading and caching.
- Collaborated on Twilio voice-AI infrastructure (**2K+ concurrent calls**) and rate limiting blocking **15K+ abusive requests/day**.

Projects

PayOps AI: Multi-Agent Payment Reconciliation | **Live Demo** | **GitHub** *TypeScript, React 19, Express 5, LangGraph | 2026*

- Developed an operator workbench with real-time updates (Socket.IO) where LangGraph and LangChain agents diagnose mismatches across 5 payment systems; risky fixes need four-eyes approval, and an eval suite passes **11/11 golden scenarios**.

MotionStudio: Browser Video Editor | **Live Demo** *React 19, Remotion, Supabase, AWS | 2026*

- Implemented a serverless render pipeline (Vercel Functions + Remotion on AWS Lambda) behind a 4-gate guard (JWT, device, quota, then invoke) so failed renders never consume quota; presigned S3 uploads and Supabase auth with RLS.
- Designed a single-mutation-point Zustand architecture powering undo/redo, autosave and cloud sync; one shared composition drives live preview, in-browser WebCodecs export and cloud rendering.

ClinRAG: Hybrid RAG over FDA Drug Labels | **GitHub** *Python, AWS Bedrock, FAISS, RAGAS | 2026*

- Created a RAG pipeline (Cohere Embed v4, FAISS plus BM25 with reciprocal rank fusion, Claude Haiku 4.5) and a RAGAS eval harness: hybrid lifted context recall from 0.75 to 0.83 on 12 hand-verified questions.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Frontend: React 19, Next.js, Redux, Zustand, React Query, Tailwind CSS, shadcn/ui, Ant Design, Core Web Vitals
Backend & Data: Node.js, Express, FastAPI, REST APIs, Socket.IO, PostgreSQL, Prisma, Redis, BullMQ, JWT/RBAC, Swagger
Cloud & DevOps: AWS (S3, SQS, SNS, Lambda, Bedrock), Vercel, Supabase, Docker, CI/CD, Dead-Letter Queues, Idempotency
AI & LLMs: Claude, Amazon Nova, Llama, Gemini, OpenAI API, RAG, LangGraph, LangChain, Structured Outputs, LLM Evals
Testing & Integrations: Jest, Supertest, Playwright, Stripe, Twilio, HubSpot, PostHog

Education

Institute of Engineering & Technology (IET), Dr. Ram Manohar Lohia Awadh University **2019 - 2023**
Bachelor of Technology in Computer Science and Engineering, 81% *Ayodhya, India*